LINEAR ALGEBRA • UNIT 5.01

C Vector Spaces & The Four Fundamental Subspaces

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 C	Q9 D	Q10 A
Q11 B	Q12 A	Q13 D	Q14 C	Q15 D
Q16 A	Q17 C	Q18 C	Q19 A	Q20 C
Q21 B	Q22 D	Q23 A	Q24 B	Q25 A

Vector Spaces & The Four Fundamental Subspaces Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C **15** ✓

D 225

WORKED SOLUTION & VERIFICATION

$\|\mathbf{u}\| = \sqrt{(-9)^2 + (-12)^2} = \sqrt{225} = 15.$

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C **29** ✓

D 41

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 29$.

3 MCQ

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

- **A** $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- **C** $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
- **D** $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

- **A** (7, 6) ✓
- **B** (6, 7)
- **C** (8, 5)
- **D** (5, 8)

WORKED SOLUTION & VERIFICATION

The transformation is defined by the matrix multiplication $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$. We are given the point $(3, 1)$, so $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Performing the matrix multiplication:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (1 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 6 + 1 \\ 3 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

- **A** 1
- **B** 0 ✓
- **C** -7
- **D** 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

We are given the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2-2(1) & -1-2(2) & 0-2(5) \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right)$$

Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right)$$

This matrix is now in row-echelon form. The corresponding system of equations is:

$x + 2y = 5$

$y = 2$

Substitute the value of y into the first equation:

$x + 2(2) = 5$

$x + 4 = 5$

$x = 1$

Therefore, the unique solution is $(1, 2)$, and the value of x is 1.

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

- **A** 13 ✓
- **B** 17
- **C** 7
- **D** 169

WORKED SOLUTION & VERIFICATION

The magnitude of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$, we have $x = 5$ and $y = -12$. Therefore, $\|\mathbf{v}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C **The distance of the point (3,4) from the origin.** ✓

D The sum of the components of the vector.

WORKED SOLUTION & VERIFICATION

The length of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by its magnitude, $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 MCQ

Consider the linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Let V be the image of T, i.e., $V = \mathrm{Im}(T)$. V is a subspace of R^3. Find the dimension of the orthogonal complement of V, denoted by $V^{\perp}$.

A 2

B 0

C 3

D **1** ✓

WORKED SOLUTION & VERIFICATION

The image of the linear transformation T is the column space of the matrix A, i.e., $V = \mathrm{Col}(A)$. The dimension of the image of T is the rank of the matrix A, i.e., $\dim(V) = \mathrm{rank}(A)$. We can find the rank of A by reducing it to row-echelon form:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow{R_2 \to R_2 - 4R_1, R_3 \to R_3 - 7R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}$$

There are two non-zero rows, so the rank of A is 2. Thus, $\dim(V) = 2$. \newline \newline We know that for a subspace V of R^n, $\dim(V) + \dim(V^{\perp}) = n$. In this case, $n = 3$ and $\dim(V) = 2$. Therefore, $\dim(V^{\perp}) = 3 - \dim(V) = 3 - 2 = 1$.

10 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ in R^2. What is the magnitude (or norm) of this vector?

- **A** 5 ✓
- **B** 7
- **C** 1
- **D** 25

WORKED SOLUTION & VERIFICATION

The magnitude (or norm) of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$, we have $x = -3$ and $y = 4$. Therefore, the magnitude is $\|\mathbf{v}\| = \sqrt{(-3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

11 MCQ

Consider the linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ represented by the matrix $A = \begin{pmatrix} 1 & 2 \\ k & 4 \end{pmatrix}$. The image of the vector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ under this transformation is $T(\mathbf{v}) = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Determine the value of the parameter k.

- **A** 2
- **B** 1 ✓
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

The linear transformation T is defined by $T(\mathbf{x}) = A\mathbf{x}$. We are given $A = \begin{pmatrix} 1 & 2 \\ k & 4 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The image of $\mathbf{v}$ is $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 1 & 2 \\ k & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 1 \\ k \cdot 1 + 4 \cdot 1 \end{pmatrix} = \begin{pmatrix} 3 \\ k + 4 \end{pmatrix}$. We are given that $T(\mathbf{v}) = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Equating the components, we have $3 = 3$ (which is consistent) and $k + 4 = 5$. Solving for k, we get $k = 5 - 4 = 1$.

12 MCQ

Consider a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \mathrm{Im}(T)$. Furthermore, let W be the subspace of R^2 defined by $W = \{(a, b) \in \mathrm{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

- **A** 1 ✓
- **B** 0
- **C** 2
- **D** Cannot be determined

WORKED SOLUTION & VERIFICATION

The linear transformation T maps R^3 to R^2. The matrix representation of T is $A = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & -2 \end{pmatrix}$. The image of T, denoted as $V = \mathrm{Im}(T)$, is the column space of A. The columns of A are $(2\ 1)$, $(-1\ 3)$, and $(1\ -2)$. Since these vectors span V, we can find a basis for V by finding the pivot columns of A or by noting that the image is a subspace of R^2. The rank of A is 2, which means $\dim(V) = 2$. Therefore, $V = \mathrm{R}^2$.

The subspace W is defined by $W = \{(a, b) \in \mathrm{R}^2 \mid a + b = 0\}$. This is the set of vectors in R^2 where the sum of the components is zero. A basis for W can be found by setting $a = -b$. For example, $(-1, 1)$ is a basis vector for W. Thus, $\dim(W) = 1$.

We are looking for the dimension of $V \cap W$. Since $V = \mathrm{R}^2$, the intersection $V \cap W$ is simply W. Therefore, $\dim(V \cap W) = \dim(W) = 1$.

13 MCQ

Consider the vector space R^2 with the standard dot product. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

A 10 | B -2

C 6 | D 2 ✓

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ in R^2 is defined as $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$. \nGiven $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -1)$, we have:\n$\mathbf{u} \cdot \mathbf{v} = (3)(2) + (4)(-1)$ \n$\mathbf{u} \cdot \mathbf{v} = 6 - 4$ \n$\mathbf{u} \cdot \mathbf{v} = 2$.

14 MCQ

Consider the vector $\mathbf{v} = (5 \ \ -12)$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 7 | B 169

C 13 ✓ | D 17

WORKED SOLUTION & VERIFICATION

$\mathbf{v} = (5 \ \ -12)$. The magnitude of a vector $\mathbf{v} = (x \ y)$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Substituting the components of $\mathbf{v}$: $\|\mathbf{v}\| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

15 MCQ

A linear map $S : \mathrm{R}^{20} \to \mathrm{R}^{28}$ has rank $\operatorname{rank}(S) = 15$. Determine the nullity $\dim(\ker S)$.

A 0 | B 20

C 8 | D 5 ✓

WORKED SOLUTION & VERIFICATION

By Rank-Nullity, $\dim(\ker S) = 20 - 15 = 5$.

16 MCQ

A planar linear transformation S scales b by 18 and y by 15 with matrix $W = \begin{pmatrix} 18 & 0 \\ 0 & 15 \end{pmatrix}$. Determine the matrix trace $\operatorname{tr}(W)$.

A 33 ✓ | B 51

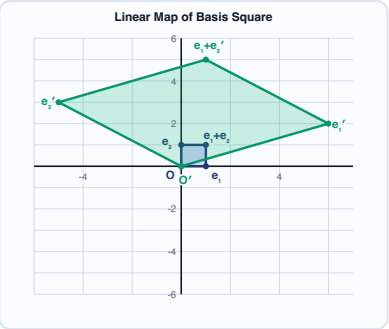
C 36 | D 18

WORKED SOLUTION & VERIFICATION

The trace of the scaling matrix is the sum of its main diagonal entries: $\operatorname{tr}(W) = 18 + 15 = 33$.

17 MCQ

For $A = \begin{bmatrix} 6 & -5 \\ 2 & 3 \end{bmatrix}$, find entry $(-3A)_{22}$.



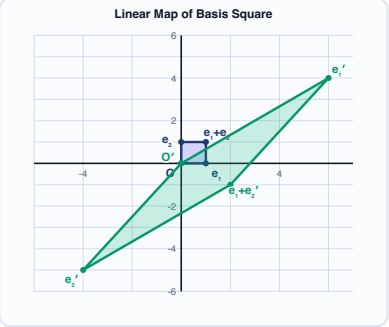
A 3	**B** 9
C -9 ✓	**D** -8

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-3A)_{22} = -3(3) = -9$. Therefore, the verified correct option is -9.

18 MCQ

For $A = \begin{bmatrix} 6 & -4 \\ 4 & -5 \end{bmatrix}$, find entry $(-2A)_{22}$.



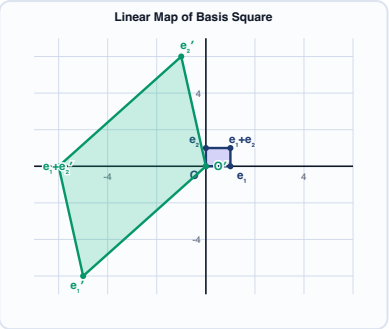
A -5	**B** -10
C 10 ✓	**D** 11

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-2A)_{22} = -2(-5) = 10$. Therefore, the verified correct option is 10.

19 MCQ

For $A = \begin{bmatrix} -5 & -1 \\ -6 & 6 \end{bmatrix}$, $B = \begin{bmatrix} -5 & -1 \\ -1 & -4 \end{bmatrix}$, find entry $(A+B)_{12}$.



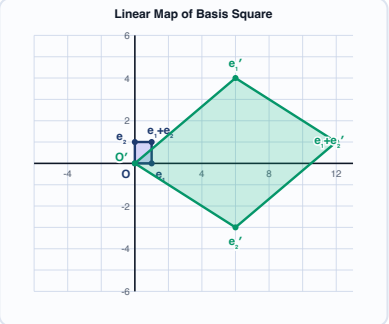
- **A** -2 ✓
- **B** -5
- **C** -1
- **D** -4

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{12} = (-1)+(-1) = -2$. Therefore, the verified correct option is -2.

20 MCQ

For $A = \begin{bmatrix} 6 & 6 \\ 4 & -3 \end{bmatrix}$, find entry $(3A)_{21}$.



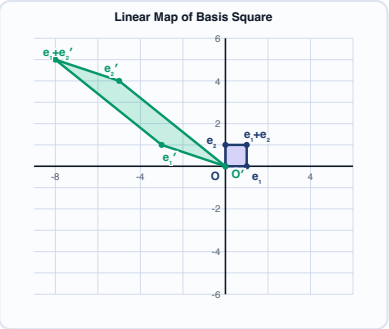
- **A** 4
- **B** -12
- **C** 12 ✓
- **D** 13

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(3A)_{21} = 3(4) = 12$. Therefore, the verified correct option is 12.

21 MCQ

For $A = \begin{bmatrix} -3 & -5 \\ 1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 4 & 6 \end{bmatrix}$, find entry $(A+B)_{12}$.



A -5

B -7 ✓

C -2

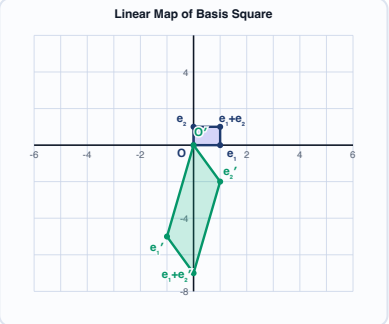
D -6

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{12} = (-5)+(-2) = -7$. Therefore, the verified correct option is -7.

22 MCQ

For $A = \begin{bmatrix} -1 & 1 \\ -5 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -5 \\ -2 & -1 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -2

B -1

C -5

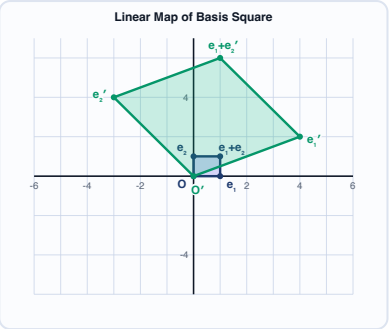
D -3 ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{22} = (-2)+(-1) = -3$. Therefore, the verified correct option is -3.

23 MCQ

For $A = \begin{bmatrix} 4 & -3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -3 \\ 4 & -1 \end{bmatrix}$, find row 1 of AB.



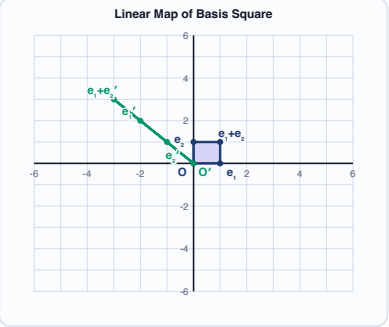
- **A** [-24, -9] ✓
- **B** [-18, -3]
- **C** [-9, -24]
- **D** [-12, 3]

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = 4(-3) + (-3)(4) = -24$, $(AB)_{12} = 4(-3) + (-3)(-1) = -9$.

24 MCQ

For $A = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ -3 & 4 \end{bmatrix}$, find row 2 of AB.



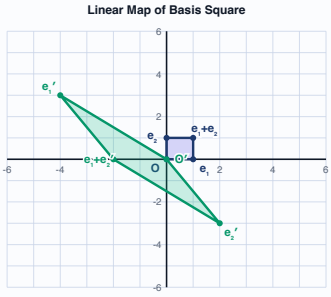
- **A** [14, 7]
- **B** [5, 2] ✓
- **C** [2, 5]
- **D** [8, 4]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = 2(4) + (1)(-3) = 5$, $(AB)_{22} = 2(-1) + (1)(4) = 2$.

25 MCQ

For $A = \begin{bmatrix} -4 & 2 \\ 3 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & -2 \end{bmatrix}$, find row 1 of AB.



A [8, 4] ✓

B [4, 8]

C [4, -4]

D [-2, 4]

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = -4(-1) + (2)(2) = 8$, $(AB)_{12} = -4(-2) + (2)(-2) = 4$.